

# Homology

## 1 Presheaves

### 1.1 Yoneda embedding

**Definition 1.1.1.** Let  $A$  be a category. A *presheaf* on  $A$  is a functor  $X : A^{\text{op}} \rightarrow \text{Set}$ . For an object  $a$  of  $A$ , write  $X_a = X(a)$  for the fibre of  $X$  at  $a$ .

A morphism  $u : a \rightarrow b$  induces the restriction map

$$u^* = X(u) : X_b \rightarrow X_a. \quad (1.1)$$

A morphism of presheaves  $f : X \rightarrow Y$  is a natural transformation. Its component maps  $f_a : X_a \rightarrow Y_a$  make the following square commute for every  $u : a \rightarrow b$ :

$$\begin{array}{ccc} X_a & \xrightarrow{f_a} & Y_a \\ \uparrow u^* & & \uparrow u^* \\ X_b & \xrightarrow{f_b} & Y_b \end{array} \quad (1.2)$$

Equivalently,  $f_a u^* = u^* f_b$ . Presheaves on  $A$  and their natural transformations form the category  $\hat{A}$ .

**Definition 1.1.2.** Yoneda embedding

$$h : A \rightarrow \hat{A}, \quad h_a = \text{hom}_A(-, a) \quad (1.3)$$

**Theorem 1.1.3.** For any presheaf  $X$  on  $A$ , there is a natural bijection

$$\begin{aligned} \text{hom}_{\hat{A}}(h_a, X) &\xrightarrow{\sim} X_a \\ \left( h_a \xrightarrow{u} X \right) &\mapsto u_a(1_a). \end{aligned} \quad (1.4)$$

For each morphism  $u : c \rightarrow a$  and section  $s \in X_a$ , define  $f_{c(u)} = u^*(s)$ . These maps are the components of a natural transformation  $f : h_a \rightarrow X$ .

## 2 Abelian categories

## 2.1 Additive and Abelian structure

**Theorem 2.1.1.** *Let  $\mathcal{A}$  be a category. We say that  $\mathcal{A}$  is additive if:*

- $\forall A, B \in \mathcal{A}$ ,  $\text{hom}(A, B)$  has the structure of an Abelian group, in such a way the composition is additive.
- Therefore, particular the set  $\text{hom}(A, b)$  is not empty as it contains the 0 element. Which is terminal and initial for  $\mathcal{A}$  unique up to isomorphism.

A consequence of this is that the identity  $\text{id}_0$  is the only element of  $\text{hom}_{\mathcal{A}}(0, 0)$ .

- In  $\mathcal{A}$ , there exists binary products and coproducts.

Fix two objects  $A, B \in \mathcal{A}$ , consider projection and inclusion maps  $p_{(-)}, i_{(-)}$ . Then  $(\text{id}_A, 0)$  and  $(\text{id}_B, 0)$

**Theorem 2.1.2.** *Let  $f : A \rightarrow B$  be a morphism in an additive category. The kernel of  $f$ , denoted as  $\ker f$  is the equalizer of  $f$  and 0. So does  $\text{coker } f$  as coequalizer of  $f$  and 0.*

**Proposition 2.1.3.** *Let  $\mathcal{A}$  be an additive category,  $f : A \rightarrow B$  a morphism. Let  $F : \mathcal{A} \rightarrow \text{Ab}$  be a functor defined by:*

$$F(X) = \ker(\text{hom}(X, A) \rightarrow \text{hom}(X, B)) \quad (2.5)$$

Let  $K = \ker f$ , fitting in universal diagram:

$$K \xrightarrow{k} A \xrightarrow{f} B \quad (2.6)$$

Then for  $X \in \mathcal{A}$ , the composition of homomorphism:

$$\text{hom}(X, K) \xrightarrow{k \circ -} \text{hom}(X, A) \xrightarrow{f \circ -} \text{hom}(X, B) \quad (2.7)$$

Is indeed 0. Which means  $\text{hom}(X, K)$  is the kernel of  $F \cong h_K$ .

**Definition 2.1.4.** Let  $\mathcal{A}$  be an additive category. We say that  $\mathcal{A}$  is an *Abelian category* if  $\mathcal{A}$  admits kernels and cokernels that identify the isomorphism on image and coimage.

$$K \xrightarrow{k} A \xrightarrow{f_1} I \xrightarrow{f_2} B \xrightarrow{l} L \quad (2.8)$$

$$0 \longrightarrow \ker f_k \longrightarrow A_k \longrightarrow \text{im } f_k \longrightarrow 0 \quad (2.9)$$

$$0 \longrightarrow A \xrightarrow{i_A} A \oplus B \xrightarrow{p_B} B \longrightarrow 0 \quad (2.10)$$

**Proposition 2.1.5.** *Let  $\mathcal{A}$  be an Abelian category,  $X \in \mathcal{A}$ , The functors  $h_X$  and  $h^X$  (with target  $\text{Ab}$ ) are both left exact.*

*Proof.*

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0 \quad (2.11)$$

Be any short exact sequence. So every morphism  $h : X \rightarrow B$  such that  $g \circ h = 0$  factors through  $A$ . Therefore:

$$0 \longrightarrow \text{hom}(X, A) \longrightarrow \text{hom}(X, B) \longrightarrow \text{hom}(X, C) \longrightarrow 0 \quad (2.12)$$

□

## 2.2 Subobjects

Given two objects  $u : U \rightarrow A$  and  $v : V \rightarrow A$  of  $A$ , if there' s a morphism  $f : U \rightarrow V$  such that  $u = v \circ f$ , then  $u$  is factored through  $v$  and  $u$  is monomorphism indicates  $f$  is monomorphism too. so  $U$  is a subobject of  $V$ . We will abuse notation and write  $U \subset V$ .

**Lemma 2.2.1.**

$$U \xrightarrow{f} V \xrightarrow{g} U \xrightarrow{u} A \quad (2.13)$$

Let  $A$  be an object of an Abelian category. Then  $U, V$  as subobjects that  $f : U \rightarrow V$  and  $g : V \rightarrow U$  are such that  $u = v \circ f$  and  $v = u \circ g$ , then above chain is enough shows that  $u = v \circ f = u \circ g \circ f$ .  $g \circ f$  is also monomorphism, shows isomorphism.

Intersection ( $\cap$ ) as pullback, for subobjects  $u : U \rightarrow A$  and  $v : V \rightarrow A$ :

$$\begin{array}{ccc} U \cap V & \xrightarrow{p_2} & V \\ p_1 \downarrow & & \downarrow v \\ U & \xrightarrow{u} & A \end{array} \quad (2.14)$$

If  $K \rightarrow U \cap V$  is such that  $K \rightarrow A$  is zero, then  $K \rightarrow U \rightarrow A$  is zero. Since  $u$  is mono,  $K \rightarrow U$  must be zero. By the same logic for  $V$ , the universal property of the pullback forces the unique map  $K \rightarrow U \cap V$  is zero. Thus  $U \cap V \rightarrow A$  is a monomorphism.

Sum ( $+$ ) as a coimage, the sum  $U + V$  is the image of the coproduct map by factor  $U \oplus V \xrightarrow{e} I \xrightarrow{m} A$ . Where  $e$  is epi and  $m$  is a mono. We define  $U + V : = I$  is the smallest subobject of  $A$  through which both  $u$  and  $v$  factor.

A morphism  $f : A \rightarrow B$  determines functors  $f_* : \text{Sub}(A) \rightarrow \text{Sub}(B)$  and  $f^* : \text{Sub}(B) \rightarrow \text{Sub}(A)$  given by direct and inverse image respectively.

$$f_*(U) \subseteq V \Leftrightarrow U \subseteq f^*(V) \quad (2.15)$$

For a subobject  $V \xrightarrow{j} B$ ,  $f^{-1}(V)$  is the pullback of  $j$  along  $f$ . Because  $V \rightarrow B$  is monomorphism,  $f^{-1}(V) \rightarrow A$  is a monomorphism. Then  $U \rightarrow A$  factors through  $f^{-1}(V)$ , then  $f(U)$  must factor through  $V$ . Reversely,  $U \xrightarrow{i} A$ ,  $f(U)$  is the image of composition  $f \circ i$ .

Assume  $f$  is mono, then for any subobject  $i : U \rightarrow A$  the composition  $f \circ i$  is mono too.

$$\begin{array}{ccc} P & \longrightarrow & U \\ j \downarrow & & f \circ i \downarrow \\ A & \longrightarrow & B \end{array} \quad (2.16)$$

The pullback  $f^{-1}(f \circ i)(U)$  retain the monomorphism and the whole diagram commutative.

$$\begin{array}{ccccccc} A & \xrightarrow{a} & B & \xrightarrow{b} & C & \xrightarrow{c} & D \\ \alpha \downarrow & & \beta \downarrow & & \gamma \downarrow & & \downarrow \delta \\ A' & \xrightarrow{a'} & B' & \xrightarrow{b'} & C' & \xrightarrow{c'} & D' \end{array} \quad (2.17)$$

### 3 Chain complexes and homology

Let  $\mathcal{A}$  be an Abelian category.

**Definition 3.1.** The category of complexes  $Kom(\mathcal{A})$  has for objects doubly infinite complexes of objects in  $\mathcal{A}$ :

$$\dots \longrightarrow A_{n-1} \longrightarrow A_n \longrightarrow A_{n+1} \longrightarrow \dots \quad (3.18)$$

$$H_n(A_\bullet) := \frac{\ker A_n \rightarrow A_{n+1}}{\text{im } A_{n-1} \rightarrow A_n} \quad (3.19)$$

If  $H_n(A_\bullet) = 0$  for all  $n$ , we say that  $A_\bullet$  is acyclic.

It's common to denote  $B_n = \text{im}(A_{n-1} \rightarrow A_n)$  for boundary and  $Z_n = \ker(A_n \rightarrow A_{n+1})$  for cycle. Then  $B_n$  is a subobject of  $Z_n$  and  $H_n(A_\bullet) = Z_n/B_n$ .

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0 \quad (3.20)$$